D4.2 Stability and change

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-d4-2.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 814–819 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Succession

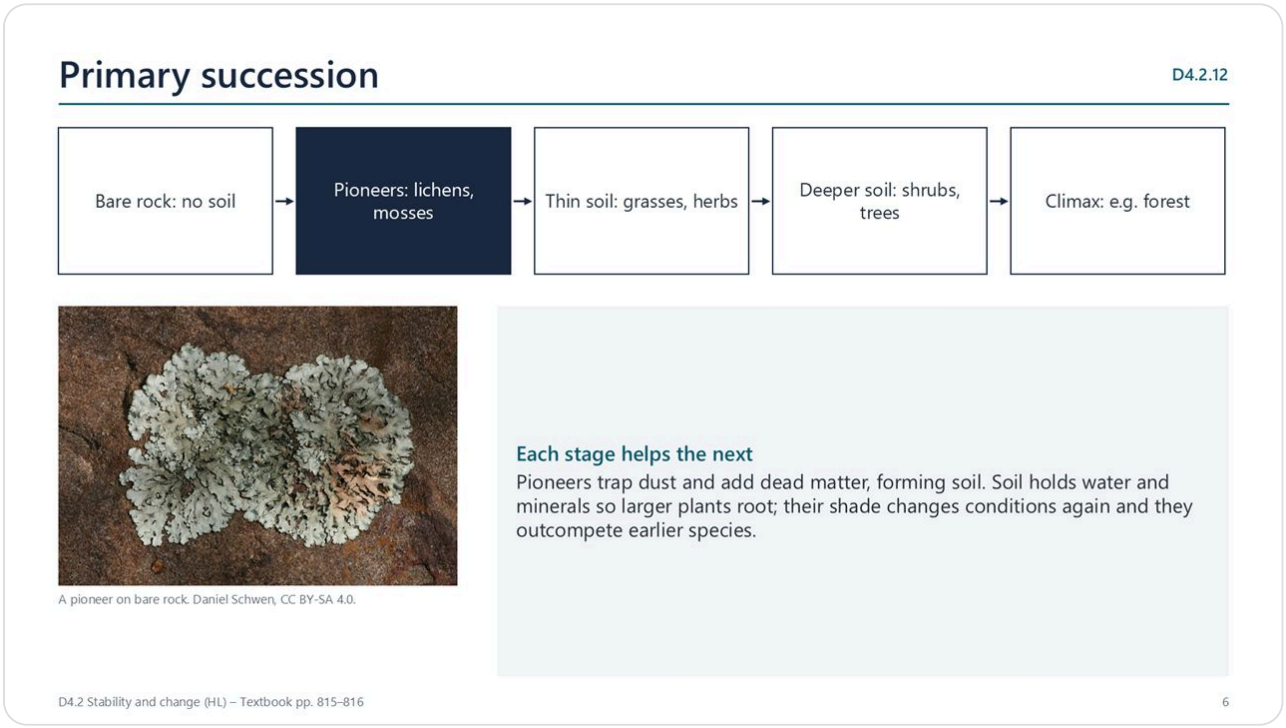
D4.2.11 What drives succession

p. 814

Succession is a directional change in the species of a community over time, driven by abiotic changes such as water, temperature and soil, and by biotic changes such as competition. Primary succession starts without soil; secondary succession starts where soil remains.

D4.2.12 Primary succession

pp. 815–816



A pioneer on bare rock. Daniel Schwen, CC BY-SA 4.0.

Pioneers such as lichens begin to form soil, which lets larger plants root. Plant size, primary production, species diversity, food-web complexity and nutrient cycling all increase.

D4.2.13 Cyclical succession

p. 817

Some ecosystems repeat a cycle of communities with no fixed climax. On some New Zealand rocky shores, barnacles, algae and then mussels settle in turn, and when the mussels detach the rock is bare again.

D4.2.14 Climax and arrested succession

pp. 818–819

A **climax community** is stable and in balance with its climate and soil. Human activity can arrest succession: livestock eat tree seedlings, and draining wetlands removes the waterlogged conditions their species need.

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